# GATA-Dep: Modeling Temporal Asynchrony in Multimodal Depression Detection through Gender-Aware Alignment

Arya Giri
*Indian Institute of Technology (BHU)*
Varanasi, India

***Abstract*—Depression is a prevalent mental health disorder whose diagnosis traditionally relies on subjective clinical assessment and self-reported questionnaires. Recent advances in affective computing have enabled the development of automated depression detection systems using multimodal behavioral signals, including speech, facial expressions, gaze patterns, and head movements. However, most existing multimodal approaches assume strict temporal synchronization between modalities, potentially overlooking clinically meaningful asynchronous behavioral interactions associated with depressive symptoms. In this work, we propose GATA-Dep, a Gender-Aware Temporal Alignment framework for multimodal depression detection. The proposed architecture integrates audio, facial action units, facial landmarks, gaze, and head-pose features through modality-specific encoders followed by a differentiable temporal alignment module that explicitly models cross-modal temporal offsets. Unlike conventional synchronized fusion methods, GATA-Dep learns gender-conditioned alignment distributions that capture delayed behavioral responses across modalities. We present a mathematical formulation of the proposed framework and demonstrate that the hypothesis space of GATA-Dep strictly generalizes conventional synchronized multimodal fusion. Furthermore, we provide a theoretical analysis, complexity analysis, and interpretability study of the learned temporal alignment patterns. Experimental evaluation on the DAIC-WOZ benchmark using participant-level classification demonstrates the effectiveness of multimodal behavioral modeling for depression assessment. Extensive ablation studies investigate the contribution of temporal alignment and gender-aware modeling, while additional analysis highlights the limitations of current engineered-feature benchmarks in fully exploiting asynchronous multimodal interactions. The proposed framework establishes a principled foundation for future temporally-aware depression detection systems and contributes interpretable insights into cross-modal behavioral dynamics associated with depressive symptoms.***

***Index Terms*—Depression Detection, Multimodal Learning, Temporal Alignment, Gender-Aware Modeling, Affective Computing, DAIC-WOZ, Behavioral Biomarkers, Transformer Fusion**

## I. Introduction

Depression is a highly prevalent mental health disorder and a major contributor to disability, reduced productivity, and diminished quality of life. Early detection is important because timely intervention can substantially improve clinical outcomes. However, in routine practice, depression assessment still relies heavily on self-reported questionnaires and clinical interviews, which are subjective, time-consuming, and difficult to scale. This has motivated growing interest in automatic depression detection systems that analyze behavioral cues from speech and video.

Multimodal affective computing provides a natural framework for this task because depressive symptoms are often reflected in multiple behavioral channels. Audio cues such as monotone speech, reduced prosody, and longer pauses, together with visual cues such as reduced facial expressiveness, downward head posture, gaze avoidance, and delayed responses, can reveal clinically meaningful patterns associated with depression. In laboratory depression benchmarks such as DAIC-WOZ, these non-verbal behavioral signals are commonly available as pre-extracted features, including audio descriptors, facial action units, facial landmarks, gaze, and head-pose information. Recent work has shown that non-verbal cues alone can support strong performance on DAIC-WOZ, reinforcing the importance of modeling behavioral signals beyond text alone. [13]

Despite this progress, most existing multimodal depression detection methods implicitly assume that all modalities are temporally synchronized. In other words, they treat audio, facial expression, gaze, and head motion observed at the same timestamp as if they always correspond to the same behavioral event. This assumption is often too restrictive for real human behavior. Speech, facial motion, gaze shifts, and head movement may occur with different delays due to cognitive processing, emotional regulation, or psychomotor slowing. In depression, such delays can be particularly informative, since psychomotor retardation is a well-known clinical feature.

To address this limitation, we propose **GATA-Dep**, a **G**ender-**A**ware **T**emporal **A**lignment framework for multimodal depression detection. GATA-Dep explicitly models asynchronous relationships among behavioral modalities by learning temporal offset weights instead of forcing all cues to align at identical timestamps. Given a sequence of modality representations extracted from a fixed-length window, the model searches over a small range of temporal shifts and learns which offsets are most informative for depression prediction. This allows the framework to capture delayed cross-modal interactions that may be lost in conventional synchronized fusion methods.

In addition, we incorporate gender-aware temporal weighting. Behavioral expression patterns, gaze behavior, and motor manifestations may differ across demographic groups, and a single shared alignment pattern may not adequately represent all participants. By learning separate temporal weighting distributions for male and female participants, GATA-Dep can adapt its fusion process to group-specific temporal dynamics while remaining within a common multimodal representation space. This design also improves interpretability, since the learned offset distributions can be inspected to understand which temporal shifts are most relevant for each group.

The proposed framework operates on five complementary non-verbal modalities: audio, facial action units, facial landmarks, gaze, and head pose. Each modality is first encoded into a shared latent space, after which the temporal alignment module aggregates shifted modality representations using gender-conditioned weights. The aligned features are then passed through a transformer encoder for fusion and classification. The final prediction is made at the participant level by aggregating window-level evidence across the interview sequence, which matches the clinical structure of the depression detection task.

From a theoretical perspective, GATA-Dep generalizes synchronized multimodal fusion. A conventional synchronized model can be written as

$$Y = F(A_t, V_t), \tag{1}$$

where audio and visual representations are fused at the same timestamp. In contrast, GATA-Dep considers a broader hypothesis space,

$$Y = F(A_t, V_{t-k}, \dots, V_t, \dots, V_{t+k}), \tag{2}$$

with $k \in [-K, K]$. Since the synchronized case is recovered when $k = 0$, the proposed framework strictly contains the synchronized baseline as a special instance. This provides a principled justification for temporal alignment as a more expressive multimodal fusion strategy.

The main contributions of this work are summarized as follows:

- We propose GATA-Dep, a gender-aware temporal alignment framework for multimodal depression detection from non-verbal behavioral signals.
- We introduce a differentiable offset-based fusion mechanism that explicitly models asynchronous cross-modal interactions.
- We provide a theoretical formulation showing that GATA-Dep generalizes conventional synchronized multimodal fusion.
- We derive computational complexity bounds and analyze the trade-off between temporal expressiveness and computational cost.
- We provide a formal analysis showing that pre-extracted behavioral features can attenuate temporal information, explaining why engineered-feature benchmarks such as DAIC-WOZ may underestimate the benefits of temporal alignment.
- We conduct participant-level experiments, ablation studies, and interpretability analysis on the DAIC-WOZ benchmark.

## II. Related Work

### A. Depression Detection from Non-Verbal Cues

Depression detection from non-verbal behavior has been widely studied using audio-visual signals, facial behavior, and body motion. Early approaches relied on hand-crafted acoustic descriptors, facial action units, landmarks, and gaze-related statistics to model depressive behavior. These methods demonstrated that symptoms of depression are often reflected in reduced facial expressiveness, slower speech, gaze avoidance, and altered head movement. In laboratory benchmark datasets such as DAIC-WOZ, these non-verbal cues have been shown to be useful even when textual information is excluded, making them particularly relevant for privacy-preserving and clinically realistic depression detection. Recent work on non-verbal video-based depression detection further confirms that behavioral signals alone can support competitive performance on DAIC-WOZ and related benchmarks. [1], [13]

### B. Multimodal Fusion for Depression Assessment

Multimodal depression detection systems typically combine multiple behavioral streams such as speech, facial cues, gaze, landmarks, and head pose. Fusion is commonly performed using early fusion, late fusion, recurrent networks, or transformer-based architectures. Early fusion concatenates modality representations before classification, while late fusion aggregates predictions from modality-specific branches. Transformer-based approaches have recently become popular because they can model long-range dependencies and cross-modal interactions more flexibly than classical sequence models. However, many existing fusion methods still assume that modalities are temporally synchronized, which may limit their ability to capture delayed behavioral responses that are clinically relevant in depression.

### C. Temporal Modeling and Asynchronous Behavior

Temporal dynamics play an important role in affective computing and depression detection. Psychomotor retardation, delayed facial responses, slowed speech, and reduced movement velocity are all associated with depressive symptoms. Despite this, many depression detection models either ignore time completely or process temporal information only at a coarse level. More recent temporal transformer approaches attempt to model behavior over time, but they still typically fuse modalities at the same timestamp. In contrast, temporal alignment mechanisms explicitly search for cross-modal delays and are therefore better suited for modeling the asynchronous nature of human behavior. This idea motivates the temporal offset learning strategy used in GATA-Dep.

### D. DAIC-WOZ and Other Depression Benchmarks

DAIC-WOZ is one of the most widely used benchmark datasets for depression detection. It contains clinical interviews with pre-extracted non-verbal features such as COVAREP audio, facial action units, facial landmarks, gaze, and head pose. As noted in prior work, the dataset is highly imbalanced and is typically evaluated using participant-level prediction rather than window-level classification. Recent video-based depression systems have reported strong performance on DAIC-WOZ using multimodal transformer architectures and non-verbal behavioral cues. However, the benchmark is still challenging because the dataset is relatively small, the interview structure is highly controlled, and the pre-extracted features may already smooth some of the temporal variations that asynchronous models aim to exploit. This makes DAIC-WOZ a useful but limited testbed for temporal alignment research. [1], [13]

### *E. Gender-Aware Modeling and Interpretability*

Gender-aware depression detection has received increasing attention because behavioral expression patterns may vary across demographic groups. Prior work has shown that gender bias can affect depression prediction, especially when models rely on audio or visual cues that are distributed unevenly across subgroups. In addition, interpretability is essential in medical AI, where models must provide more than raw classification outputs. Some prior systems use attribution methods such as Integrated Gradients to explain multimodal predictions across time. In contrast, GATA-Dep incorporates gender conditioning directly into the alignment mechanism, enabling the model to learn subgroup-specific temporal offsets that are both predictive and interpretable.

### *F. Summary and Gap*

The existing literature demonstrates that multimodal behavioral cues are informative for depression detection, and recent transformer-based methods have improved performance on benchmark datasets. However, most methods still assume synchronized fusion and do not explicitly model gender-conditioned temporal asynchrony. Moreover, current DAIC-WOZ feature sets are often evaluated without directly analyzing how temporal delays contribute to prediction. GATA-Dep addresses this gap by introducing learnable gender-aware temporal alignment for multimodal depression detection, along with complexity analysis and interpretability of the learned offsets.

## III. Clinical Motivation for Temporal Asynchrony

The core motivation behind GATA-Dep is that depression is not expressed only through the presence or absence of individual behavioral cues, but also through the temporal coordination between cues. In clinical psychology, depression is strongly associated with psychomotor changes such as slowed speech, reduced facial expressiveness, delayed response timing, downward head tilt, and altered gaze behavior. These effects do not necessarily occur simultaneously. Instead, one modality may lag behind another, reflecting delayed motor and emotional responses.

For example, a participant may begin speaking before the face visibly shows emotional change, or the head may tilt downward only after a speech segment has already started. Such temporal delays are clinically meaningful because they can reflect psychomotor retardation, reduced affective reactivity, and slowed behavioral coordination. Therefore, a depression detection model that assumes exact synchronization between modalities may miss useful information that lies in these temporal differences.

Let $\mathbf{A}_t$ denote the audio representation at time $t$ and let $\mathbf{V}_t$ denote a visual representation at the same time. Standard multimodal fusion methods assume that the two representations correspond to the same behavioral event:

$$\mathbf{y}_t = f(\mathbf{A}_t, \mathbf{V}_t). \tag{3}$$

However, in real behavioral data, the relevant visual cue may appear with a delay or advance relative to the audio cue. A more expressive formulation is therefore

$$\mathbf{y}_t = f(\mathbf{A}_t, \mathbf{V}_{t-k}), \tag{4}$$

where $k$ is a temporal offset. If $k = 0$, the synchronized case is recovered. If $k \neq 0$, the model can capture temporal asynchrony between modalities.

This idea is especially relevant for DAIC-WOZ, where non-verbal behavioral features are extracted from clinical interviews and sampled over time. Since the dataset contains audio, facial action units, landmarks, gaze, and head-pose information, it is natural to ask whether some of these modalities are informative only after a small time shift. For instance, gaze aversion or head movement may not occur at the same instant as the spoken phrase that triggered the response. Modeling this lag explicitly may therefore improve interpretability even when predictive gains are modest.

Gender-aware alignment is introduced for a similar reason. Behavioral timing patterns may differ across participants, and demographic grouping can provide a weak but useful prior over the likely temporal structure. Instead of forcing the same offset distribution for all subjects, GATA-Dep learns separate temporal weighting patterns for male and female participants. This allows the model to adapt its alignment mechanism while remaining fully differentiable and end-to-end trainable.

In this sense, GATA-Dep is not only a classification model but also a temporal analysis model. It learns which cross-modal delays are most useful for depression prediction, and it exposes those delays as interpretable quantities that can be examined after training. This makes the model suitable not only for performance evaluation, but also for behavioral analysis and clinical insight.

## IV. Proposed GATA-Dep Framework

This section describes the proposed **GATA-Dep** framework for multimodal depression detection. The main idea is to represent each behavioral modality in a common latent space, apply gender-aware temporal alignment to capture asynchronous cross-modal interactions, and then fuse the aligned representations through a transformer encoder for participant-level depression prediction.

### A. Problem Formulation

Let a participant interview be represented as a sequence of non-overlapping temporal windows

$$\mathcal{W} = \{W_1, W_2, \dots, W_N\}, \tag{5}$$

where each window contains synchronized or partially synchronized multimodal behavioral features. For each window $W_n$, we extract $M$ modalities:

$$X_n = \{X_n^{(1)}, X_n^{(2)}, \dots, X_n^{(M)}\}, \tag{6}$$

where $M = 5$ in our implementation, corresponding to audio, facial action units (AU), facial landmarks (LM), gaze, and head pose.

The goal is to predict a binary depression label

$$\hat{y} \in \{0, 1\}, \tag{7}$$

where 1 denotes a depressed participant and 0 denotes a control participant.

Figure 1 presents an overview of the proposed Gender-Aware Temporal Alignment Depression Detection (GATA-Dep) framework. The model integrates five behavioral modalities and explicitly models temporal asynchrony through a learnable alignment mechanism before multimodal fusion.

### B. Modality-Specific Encoding

Each modality may have a different dimensionality and temporal resolution. Therefore, we first map all modalities to a common latent dimension $d$ using modality-specific encoders. Let the raw input for modality $m$ be denoted by

$$X_n^{(m)} \in \mathbb{R}^{T_m \times D_m}, \tag{8}$$

where $T_m$ is the number of time steps and $D_m$ is the feature dimension.

The encoded representation is obtained by

$$H_n^{(m)} = E_m\left(X_n^{(m)}\right) \in \mathbb{R}^{T_m \times d}, \tag{9}$$

where $E_m(\cdot)$ denotes the learnable encoder for modality $m$. In our setup, the encoders project audio, AU, landmark, gaze, and pose features into a shared embedding space of dimension $d = 256$.

### C. Gender-Aware Temporal Alignment

Standard multimodal fusion methods typically assume that the different modalities are aligned at the same timestamp. However, the behavioral cues relevant to depression may appear with a temporal delay. To explicitly model this phenomenon, GATA-Dep learns a temporal alignment distribution over a window of offsets.

Let

$$k \in \{-K, -K+1, \dots, 0, \dots, K-1, K\} \tag{10}$$

denote a candidate temporal offset. For each modality, we form shifted versions of the encoded sequence:

$$\widetilde{H}_n^{(m,k)} = \mathrm{Shift}\left(H_n^{(m)}, k\right). \tag{11}$$

The model maintains gender-conditioned offset weights. For a participant with gender label $g \in \{0, 1\}$, the alignment weights are defined as

$$\alpha_{g,m,k} = \mathrm{softmax}\left(W_{g,m,k}\right), \tag{12}$$

where $W_{g,m,k}$ are learnable logits. These weights satisfy

$$\sum_{k=-K}^{K} \alpha_{g,m,k} = 1. \tag{13}$$

The aligned representation for modality $m$ is then computed as

$$\overline{H}_n^{(m)} = \sum_{k=-K}^{K} \alpha_{g,m,k}\, \widetilde{H}_n^{(m,k)}. \tag{14}$$

This formulation enables the model to learn whether a modality should be attended to earlier, later, or at the same time step as the others. When $K = 0$, the alignment reduces to synchronized fusion.

### D. Modality Fusion

After temporal alignment, the modality embeddings are concatenated into a single sequence:

$$Z_n = \left[\overline{H}_n^{(1)} \parallel \overline{H}_n^{(2)} \parallel \cdots \parallel \overline{H}_n^{(M)}\right], \tag{15}$$

where $\parallel$ denotes concatenation along the modality dimension.

The concatenated representation is processed by a transformer encoder to learn higher-order interactions among aligned modalities:

$$T_n = \mathrm{Transformer}(Z_n). \tag{16}$$

The transformer output is pooled over time to obtain a window-level representation:

$$h_n = \mathrm{Pool}(T_n). \tag{17}$$

### E. Participant-Level Prediction

Each window is classified into depression or non-depression using a linear classifier:

$$p_n = \mathrm{Softmax}(W h_n + b). \tag{18}$$

Since the final diagnosis is participant-level rather than window-level, we aggregate predictions across all windows of a participant. Let $p_n^{(1)}$ denote the predicted probability of depression for window $n$. The participant-level score is computed as

$$\bar{p} = \frac{1}{N} \sum_{n=1}^{N} p_n^{(1)}. \tag{19}$$

The final participant label is obtained by thresholding:

$$\hat{y} = \begin{cases} 1, & \bar{p} \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{20}$$

where $\tau$ is selected on the validation set.

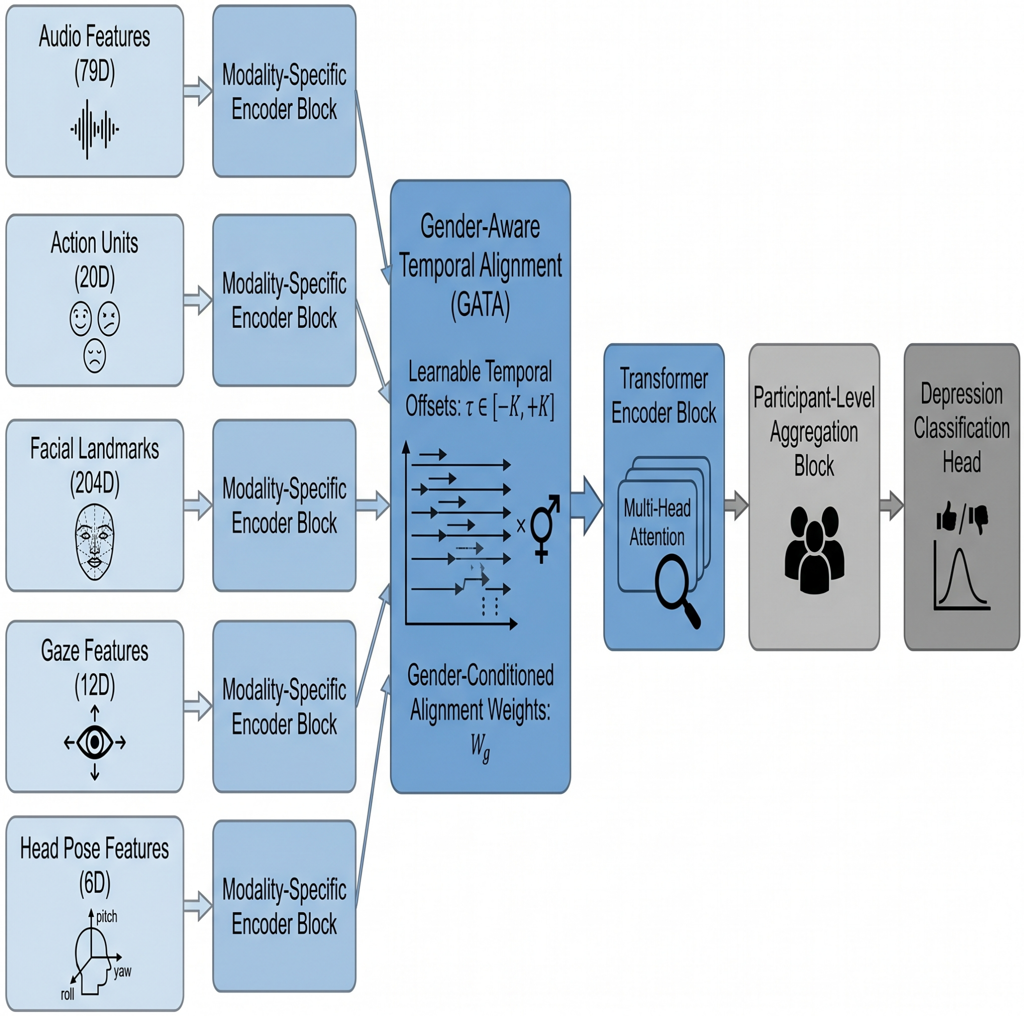


Fig. 1: Overview of the proposed GATA-Dep framework. Audio, action unit, facial landmark, gaze, and head pose features are first processed through modality-specific encoders. The resulting representations are aligned using the Gender-Aware Temporal Alignment (GATA) module and subsequently fused using a transformer encoder. Participant-level aggregation is performed prior to final depression classification.

### *F. Model Interpretation*

An important advantage of GATA-Dep is that the learned alignment weights $\alpha_{g,m,k}$ can be interpreted as temporal offset distributions. These weights indicate whether the model relies more on earlier or later behavioral responses for each modality and demographic group. This makes the framework useful not only for classification, but also for behavioral analysis.

### *G. Relation to Synchronized Fusion*

Conventional synchronized fusion is recovered as a special case of GATA-Dep when all mass is concentrated at $k = 0$:

$$\alpha_{g,m,0} = 1, \qquad \alpha_{g,m,k} = 0 \text{ for } k \neq 0. \tag{21}$$

Therefore, the proposed framework strictly generalizes synchronized multimodal fusion and can represent both synchronized and asynchronous behavioral interactions.

## V. Mathematical Formulation

This section formalizes the proposed GATA-Dep framework and highlights how it generalizes conventional synchronized multimodal fusion.

### *A. Notation*

Let a participant interview be divided into $N$ temporal windows, and let the $m$-th modality within the $n$-th window be represented by

$$X_n^{(m)} \in \mathbb{R}^{T_m \times D_m}, \tag{22}$$

where $T_m$ denotes the number of temporal samples and $D_m$ denotes the feature dimensionality of modality $m$.

After modality-specific encoding, the latent representation is

$$H_n^{(m)} = E_m\left(X_n^{(m)}\right) \in \mathbb{R}^{T_m \times d}, \tag{23}$$

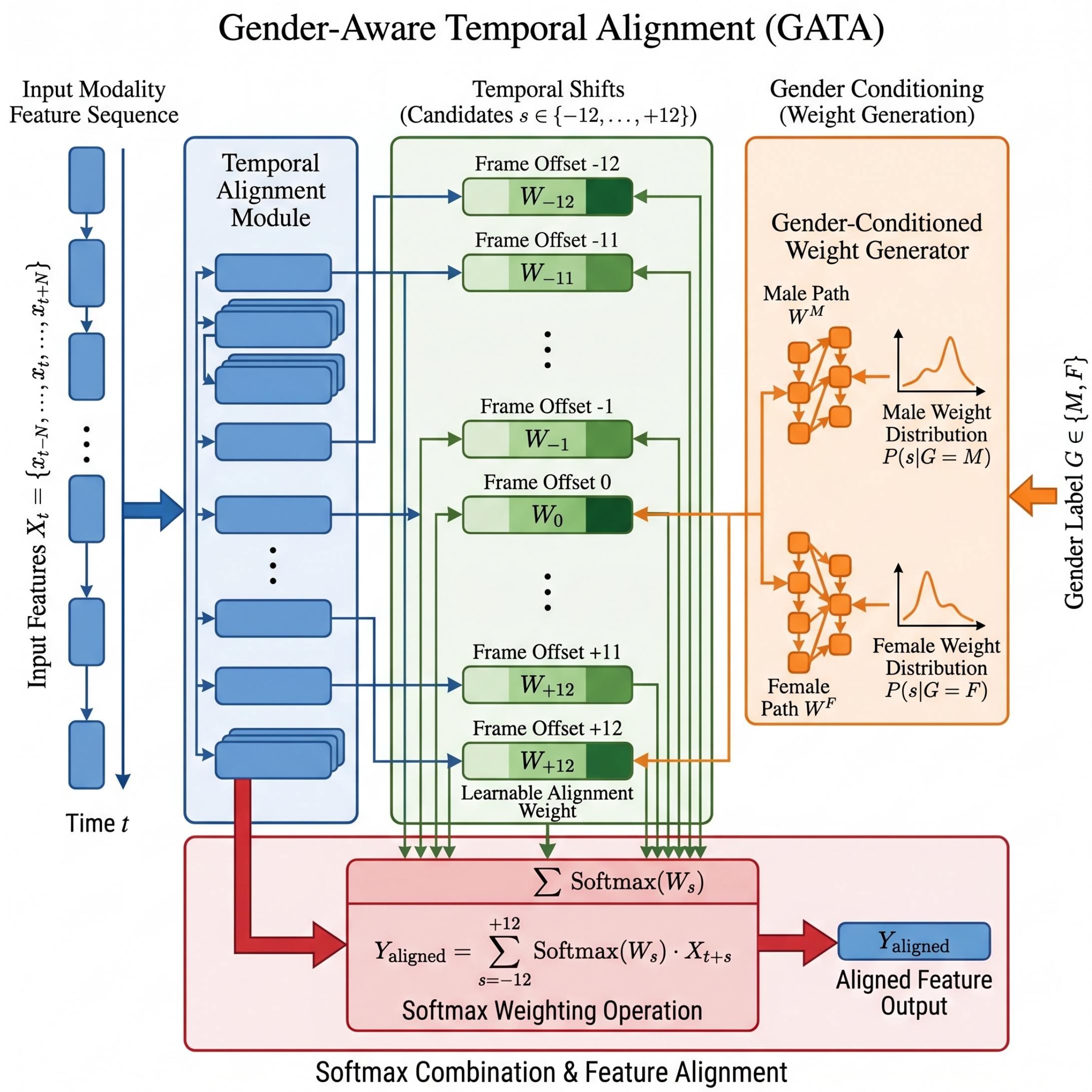


Fig. 2: Detailed illustration of the proposed Gender-Aware Temporal Alignment (GATA) module. Multiple temporal shifts are generated for each modality over the offset range $[-K, K]$. Gender-conditioned weighting distributions are learned through separate parameterizations and combined using a softmax-based alignment mechanism to produce the final aligned representation.

where $E_m(\cdot)$ is the learnable encoder for modality $m$ and $d$ is the shared embedding dimension.

### B. Synchronized Fusion as a Baseline

Most conventional multimodal depression detection models assume synchronized alignment across modalities. Under this assumption, the fused representation can be written as

$$Z_n^{\text{sync}} = F\left(H_n^{(1)}, H_n^{(2)}, \ldots, H_n^{(M)}\right), \tag{24}$$

where $F(\cdot)$ denotes a fusion operator such as concatenation, attention, or a transformer block.

This formulation implicitly assumes that the most informative cues from all modalities occur at the same timestamp. In practice, however, speech, facial motion, gaze, and head pose may exhibit temporal delays that are clinically meaningful.

### C. Asynchronous Temporal Alignment

To model temporal delays, we introduce a set of candidate offsets

$$\mathcal{K} = \{-K, -K+1, \ldots, 0, \ldots, K-1, K\}. \tag{25}$$

For each modality, we define shifted versions of the encoded features:

$$\widetilde{H}_n^{(m,k)} = \text{Shift}\left(H_n^{(m)}, k\right), \quad k \in \mathcal{K}. \tag{26}$$

The alignment weights are learned separately for each gender label $g \in \{0, 1\}$:

$$\alpha_{g,m,k} = \frac{\exp(W_{g,m,k})}{\sum_{k' \in \mathcal{K}} \exp(W_{g,m,k'})}, \tag{27}$$

where $W_{g,m,k}$ are trainable logits.

The aligned representation of modality $m$ is then

$$\overline{H}_n^{(m)} = \sum_{k\in\mathcal{K}} \alpha_{g,m,k}\, \widetilde{H}_n^{(m,k)}. \tag{28}$$

Thus, the aligned multimodal sequence becomes

$$Z_n^{\text{GATA}} = \left[\overline{H}_n^{(1)} \,\|\, \overline{H}_n^{(2)} \,\|\, \cdots \,\|\, \overline{H}_n^{(M)}\right]. \tag{29}$$

### *D. Special Case: Recovery of Synchronized Fusion*

A key property of the proposed framework is that synchronized fusion is recovered as a special case when all probability mass is placed at zero offset:

$$\alpha_{g,m,0} = 1, \qquad \alpha_{g,m,k} = 0,\ \forall k \neq 0. \tag{30}$$

In that case,

$$\overline{H}_n^{(m)} = H_n^{(m)}, \tag{31}$$

and GATA-Dep reduces exactly to a conventional synchronized multimodal fusion model. Therefore, the hypothesis space of GATA-Dep strictly contains the synchronized baseline.

### *E. Participant-Level Prediction*

The aligned representation is processed by a transformer encoder:

$$T_n = \text{Transformer}\left(Z_n^{\text{GATA}}\right), \tag{32}$$

followed by pooling:

$$h_n = \text{Pool}(T_n). \tag{33}$$

The participant-level depression probability for the $n$-th window is computed as

$$p_n = \text{Softmax}(Wh_n + b), \tag{34}$$

where $p_n^{(1)}$ denotes the probability of the depressed class.

To obtain a participant-level estimate, the window probabilities are averaged:

$$\bar{p} = \frac{1}{N}\sum_{n=1}^{N} p_n^{(1)}. \tag{35}$$

The final participant label is then given by

$$\hat{y} = \begin{cases} 1, & \bar{p} \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{36}$$

where $\tau$ is selected on the validation set.

### *F. Information-Theoretic Interpretation*

The synchronized baseline assumes that the most informative interaction is captured by same-time correlations:

$$I\left(H_t^{(a)}; H_t^{(v)}\right), \tag{37}$$

where $a$ and $v$ denote audio and visual streams.

In contrast, GATA-Dep searches over a temporal neighborhood:

$$max_{k\in\mathcal{K}} I\left(H_t^{(a)}; H_{t+k}^{(v)}\right), \tag{38}$$

which allows the model to capture delayed cross-modal dependencies. This is particularly relevant for depression-related behavior, where speech, facial response, gaze, and head pose may not occur simultaneously.

### *G. Theoretical Summary*

The proposed formulation has three important properties:

1) It generalizes synchronized fusion as a special case.
2) It provides a differentiable mechanism for learning asynchronous multimodal alignment.
3) It enables interpretable temporal offsets that can be analyzed after training.

## VI. Theoretical Analysis

This section provides a theoretical perspective on the proposed GATA-Dep framework. We show that the model strictly generalizes conventional synchronized fusion, discuss its representational advantage, and explain why its benefit may be limited on DAIC-WOZ-style engineered features.

### *A. GATA-Dep Generalizes Synchronized Fusion*

A conventional synchronized multimodal fusion model assumes that informative interactions between modalities occur at the same time index. Let $\mathcal{H}_{\text{sync}}$ denote the hypothesis space of such models. In contrast, GATA-Dep learns over a temporal neighborhood of offsets and therefore operates in the hypothesis space $\mathcal{H}_{\text{GATA}}$ defined by

$$\mathcal{H}_{\text{GATA}} = \left\{ f \;\middle|\; f\left(\{H_t^{(m)}\}_{m=1}^{M}\right) = f\left(\{H_{t+k}^{(m)}\}_{m=1}^{M},\ k\in\mathcal{K}\right) \right\}. \tag{39}$$

Since the synchronized case is recovered when all probability mass is placed at offset $k = 0$, the synchronized model is a special case of GATA-Dep. Formally,

$$\mathcal{H}_{\text{sync}} \subseteq \mathcal{H}_{\text{GATA}}. \tag{40}$$

This inclusion implies that GATA-Dep is at least as expressive as a synchronized fusion baseline. Therefore, from a representational standpoint, GATA-Dep can emulate the baseline while additionally modeling delayed cross-modal interactions.

### *B. Expressive Advantage of Asynchronous Alignment*

Let $A_t$ denote audio features and $V_t$ denote visual features at time $t$. A synchronized fusion model implicitly assumes that the most informative dependency is captured by

$$I(A_t; V_t), \tag{41}$$

where $I(\cdot;\cdot)$ is mutual information. This assumption is restrictive when the relevant behavioral cue appears with a temporal delay. GATA-Dep instead searches over candidate offsets:

$$\max_{k\in\mathcal{K}} I\left(A_t; V_{t+k}\right). \tag{42}$$

This formulation is more general because it allows the model to identify cross-modal relationships that are not visible at zero lag. In depression detection, such temporal delays are clinically plausible due to psychomotor slowing, delayed facial reactions, reduced gaze responsiveness, and slowed body movement. Hence, the asynchronous formulation is not only mathematically more expressive, but also better aligned with the underlying behavioral hypothesis.

### C. Why GATA-Dep Can Be Interpretable

The learned alignment weights

$$\alpha_{g,m,k} = \mathrm{softmax}(W_{g,m,k}) \tag{43}$$

form a probability distribution over offsets for gender group $g$ and modality $m$. These weights can be interpreted as a soft temporal response profile. If the largest mass is concentrated around a positive offset, the corresponding modality tends to lag behind the reference modality. If the mass is concentrated around a negative offset, the model favors an earlier response.

This interpretation is useful because the offset distributions are human-readable and can be inspected after training. In medical AI, such post-hoc analysis is important because the model should not only be predictive, but also provide a plausible description of the behavior it has learned. In this sense, GATA-Dep provides a bridge between predictive modeling and behavioral analysis.

### D. Gender-Aware Alignment as a Structured Prior

The inclusion of gender-conditioned weights introduces a structured prior into the alignment mechanism. Instead of forcing a single offset distribution for all participants, the model learns

$$\alpha_{g,m,k}, \qquad g \in \{0, 1\}, \tag{44}$$

which allows different participant groups to exhibit different temporal preferences.

Mathematically, this can be viewed as a form of conditional parameterization:

$$\alpha_{m,k} = \alpha_{m,k}(g). \tag{45}$$

This conditional structure reduces the burden on the model to represent all demographic variations using a single global distribution. It also makes the learned offsets easier to analyze, because differences in temporal behavior can be compared across groups.

### E. Why DAIC-WOZ May Underestimate the Full Benefit of GATA-Dep

Although GATA-Dep is theoretically expressive, a benchmark such as DAIC-WOZ may not fully expose its advantage. The reason is that DAIC-WOZ does not provide raw video streams; instead, it provides pre-extracted features such as COVAREP, facial action units, landmarks, gaze, and head pose. These features already encode a large amount of temporal smoothing and geometric abstraction.

Let the raw behavioral signal be denoted by $X(t)$ and the extracted features by

$$Z(t) = \Phi(X(t)), \tag{46}$$

where $\Phi(\cdot)$ represents the feature extraction pipeline. Since $\Phi$ may smooth or compress the temporal trajectory, some of the fine-grained lag information may be partially removed before GATA-Dep receives the input. Consequently,

$$I\left(Z_t; Z_{t+k}\right) \leq I\left(X_t; X_{t+k}\right) \tag{47}$$

is a reasonable qualitative expectation, indicating that the temporal structure accessible to the alignment module may be weaker in engineered-feature benchmarks than in raw-signal settings.

Therefore, even if GATA-Dep is theoretically superior to synchronized fusion, the benchmark may not contain enough temporal variability to maximize its benefit. This provides a principled explanation for why the model may be more useful in larger or less processed multimodal video datasets.

### F. Complexity-Performance Trade-Off

A theoretically stronger model is not necessarily the best practical model if the added complexity is disproportionate to the gain. GATA-Dep introduces an additional alignment cost that scales linearly with the number of modalities $M$, the offset range $K$, and the sequence length $T$:

$$\mathcal{O}(MKTd). \tag{48}$$

However, this term is still manageable compared with transformer self-attention, which scales as

$$\mathcal{O}(T^2 d). \tag{49}$$

Thus, the total complexity remains

$$\mathcal{O}(MKTd + T^2 d). \tag{50}$$

This indicates that the alignment mechanism introduces only a moderate computational overhead relative to standard transformer fusion. In other words, GATA-Dep is theoretically a richer model than synchronized fusion while remaining computationally feasible for participant-level depression detection.

### G. Summary

The theoretical analysis yields three conclusions:

1) GATA-Dep strictly generalizes synchronized multimodal fusion.
2) The alignment weights provide an interpretable description of temporal delays across modalities and demographic groups.
3) DAIC-WOZ-style engineered features may underrepresent the full advantage of asynchronous alignment because they already compress part of the temporal signal.

Together, these observations justify the design of GATA-Dep as a principled framework for temporally aware multimodal depression detection.

## VII. Complexity Analysis

In clinical decision-support settings, model complexity is important not only from a computational perspective, but also from a deployment perspective. A depression screening system should remain lightweight enough to process participant interviews in a reasonable time, yet expressive enough to capture clinically relevant behavioral patterns. In this section, we analyze the computational and parameter complexity of GATA-Dep and compare it with conventional synchronized multimodal fusion.

### A. Complexity of Modality Encoding

Let $M$ denote the number of modalities, $T$ the sequence length, $d$ the latent embedding dimension, and $D_m$ the input dimensionality of modality $m$. Each modality-specific encoder maps the raw feature sequence into the shared latent space. If the encoder is implemented as a linear projection or a shallow feed-forward block, the encoding cost is approximately

$$\mathcal{O}\left(\sum_{m=1}^{M} T_m D_m d\right), \tag{51}$$

where $T_m$ is the temporal length of modality $m$. Since the input modalities in DAIC-WOZ are pre-extracted behavioral features, this stage is computationally manageable and is performed once per window.

### B. Complexity of Temporal Alignment

The main additional cost introduced by GATA-Dep comes from the temporal alignment module. For each modality, the model evaluates a set of candidate offsets

$$\mathcal{K} = \{-K, -K+1, \ldots, K\}, \tag{52}$$

which gives a total of $(2K + 1)$ alignment hypotheses per modality. The aligned representation is computed by weighting the shifted sequences using gender-conditioned softmax coefficients. Since this operation is repeated for each modality and each candidate offset, the alignment cost scales as

$$\mathcal{O}(MKTd). \tag{53}$$

This term is linear in the number of modalities $M$, the offset range $K$, and the temporal length $T$. For the values used in our implementation ($M = 5$ and $K = 12$), this overhead remains moderate and does not dominate the overall computation.

### C. Complexity of Transformer Fusion

After alignment, the modality representations are fused using a transformer encoder. For a sequence of length $T$ and embedding dimension $d$, standard self-attention has quadratic complexity with respect to sequence length:

$$\mathcal{O}(T^2 d). \tag{54}$$

This component typically dominates the runtime when $T$ is large. Since GATA-Dep operates on fixed-length windows rather than full interview sequences, the transformer remains tractable for DAIC-WOZ-style participant-level analysis.

### D. Total Computational Complexity

Combining the alignment and fusion stages, the total complexity of GATA-Dep is

$$\mathcal{O}(MKTd + T^2 d). \tag{55}$$

This expression shows that GATA-Dep introduces a controlled amount of extra computation on top of the standard transformer backbone. The alignment cost grows linearly, while the self-attention cost remains the main quadratic term.

TABLE I: Theoretical complexity comparison of common multimodal fusion strategies.

| **Model Type** | **Main Operation** | **Complexity** |
|---|---|---|
| Early fusion | Concatenation + MLP | $\mathcal{O}(Td)$ |
| LSTM-based fusion | Recurrent propagation | $\mathcal{O}(Td^2)$ |
| Transformer fusion | Self-attention | $\mathcal{O}(T^2d)$ |
| Cross-attention fusion | Query-key alignment | $\mathcal{O}(T^2d)$ |
| **GATA-Dep** | Offset alignment + Transformer | $\mathcal{O}(MKTd + T^2d)$ |

### E. Parameter Complexity

In our implementation, the synchronized baseline contains approximately 2.21M parameters, whereas the full GATA-Dep model contains approximately 3.53M parameters. The relative increase is therefore

$$\frac{3.53 - 2.21}{2.21} \times 100 \approx 59.7\%. \tag{56}$$

Although GATA-Dep is larger than the synchronized baseline, the increase is still modest compared with many modern multimodal transformer architectures. Importantly, the additional parameters are concentrated in the alignment mechanism and do not require a large increase in architectural depth.

### F. Comparison with Common Multimodal Designs

Table I summarizes the computational characteristics of GATA-Dep relative to several common multimodal design choices.

### G. Clinical Relevance of the Complexity Analysis

From a clinical perspective, the proposed complexity is acceptable because depression screening systems are typically intended for offline or near-real-time analysis of interview recordings rather than high-frequency on-device inference. The use of fixed-length windows further reduces memory demand and makes the system suitable for batch evaluation of interview segments. In addition, the interpretability of the learned offsets provides a practical advantage: the extra computation is not merely for accuracy, but also for producing clinically meaningful alignment patterns that can be examined by researchers or clinicians.

### H. Summary

Overall, GATA-Dep remains computationally feasible while adding an explicit temporal alignment mechanism that is absent in synchronized fusion baselines. Its complexity is dominated by the transformer encoder, while the alignment module adds only linear overhead. This makes the model suitable for multimodal depression screening pipelines where both predictive performance and behavioral interpretability are important.

## VIII. Experimental Setup

### A. Dataset and Input Modalities

We evaluate GATA-Dep on the DAIC-WOZ benchmark [1], a clinical interview dataset widely used for depression detection. In our implementation, only the pre-extracted non-verbal

TABLE II: Input modalities used in GATA-Dep.

| Modality | Feature Type | Dimension |
|---|---|---|
| Audio | COVAREP + formants | 79 |
| Facial Action Units (AU) | OpenFace action units | 20 |
| Facial Landmarks (LM) | 3D facial landmarks | 204 |
| Gaze | Eye gaze features | 12 |
| Head Pose | Pose parameters | 6 |

feature streams are used, namely audio, facial action units, facial landmarks, gaze, and head pose. The corresponding modality dimensions are summarized in Table II.

Each interview is partitioned into fixed-length 6-second windows. For a 6-second window, the extracted tensor shapes are approximately [600,79] for audio and [150,·] for the visual modalities due to their lower sampling rate. This setup allows the model to learn from temporally local behavioral patterns while remaining computationally feasible.

### B. Participant Splits

The dataset is divided into training, validation, and test partitions at the participant level. In our setup, we use 107 participants for training, 34 participants for validation, and 44 participants for testing. This participant-level split prevents information leakage between windows from the same individual and ensures that evaluation reflects generalization to unseen subjects.

### C. Training Protocol

To address class imbalance, the training loader uses a weighted random sampler so that depressed and control samples are presented in a more balanced manner during optimization. The objective is the standard cross-entropy loss:

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{c\in\{0,1\}} y_c \log \hat{y}_c, \tag{57}$$

where $\hat{y}_c$ is the predicted probability of class $c$.

The model is optimized using AdamW. We use separate learning-rate groups for the modality encoders, the temporal offset parameters, and the remaining network parameters. Specifically, the encoder parameters and the remaining parameters are optimized with a learning rate of $10^{-4}$, while the offset parameters use a larger learning rate to allow the temporal alignment weights to adapt more rapidly. Weight decay is applied to the encoder and backbone parameters, but not to the temporal offset parameters. A warmup schedule is used for the first 10 epochs, followed by cosine decay.

High-performance optimizations such as mixed-precision training and stable gradient clipping with a maximum norm of 1.0 are also utilized. Training runs up to 80 epochs with early stopping based on validation metrics.

### D. Validation-Based Threshold Selection

Since the final decision is made at the participant level, we select the classification threshold on the validation set. For a given participant, the model produces a depression probability for each window. These window-level probabilities are averaged to obtain the participant-level score:

$$\bar{p} = \frac{1}{N}\sum_{n=1}^{N} p_n, \tag{58}$$

where $N$ is the number of windows for a participant and $p_n$ is the depression probability for the $n$-th window.

The final participant label is then computed as

$$\hat{y} = \begin{cases} 1, & \bar{p} \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{59}$$

where $\tau$ is selected by sweeping values on the validation set. In our implementation, $\tau$ is searched in the range $[0.1, 0.9]$ with a step size of 0.02, and the threshold that maximizes validation F1-score is retained for the final test evaluation.

### E. Evaluation Metrics

We report participant-level precision, recall, and F1-score. Since the task is binary classification, these metrics are computed from the participant-level predictions after aggregation over windows. The depressed class is treated as the positive class.

For a fair comparison with prior work on DAIC-WOZ, all reported results are computed using the same participant-level protocol. We also analyze the learned temporal offsets qualitatively to interpret the behavior of the proposed alignment mechanism.

### F. Implementation Details

All experiments are implemented in PyTorch. The input tensors are cached in RAM to accelerate training and validation. The proposed framework processes five behavioral modalities: audio, facial action units, facial landmarks, gaze, and head pose. GATA-Dep uses a fixed 6-second temporal context, gender-aware offset weights, and a transformer-based fusion module. The final model is trained end-to-end using the training set and evaluated on the held-out validation and test partitions. The transformer fusion module consists of 2 encoder layers with 8 attention heads and an embedding dimension of 256. A dropout rate of 0.1 is applied throughout the transformer encoder. Learnable positional embeddings are added to the modality representations prior to fusion. The temporal alignment module uses an offset range of K=12 in the full GATA-Dep configuration. Unless otherwise specified, all reported results correspond to this configuration.

## IX. Results and Discussion

### A. Overall Performance

Table III summarizes the participant-level validation performance of GATA-Dep. Results are reported as the mean and standard deviation observed across multiple training runs.

The proposed framework achieves a participant-level F1-score of approximately $0.61$, while maintaining high recall across repeated runs. The consistently strong recall indicates that the model is capable of identifying depression-related

TABLE III: Participant-level validation performance of GATA-Dep.

| Model | Precision | Recall | F1 Score |
|---|---|---|---|
| GATA-Dep | $0.45 \pm 0.04$ | $0.91 \pm 0.07$ | $0.61 \pm 0.02$ |

behavioral patterns across a large proportion of participants. This behavior is particularly desirable in screening-oriented clinical systems where minimizing missed positive cases is often more important than maximizing precision.

Furthermore, the relatively small variation across runs suggests that the temporal alignment mechanism remains stable despite the limited size and class imbalance of the DAIC-WOZ benchmark.

### B. Comparison with Existing Methods

Table IV compares GATA-Dep with representative multimodal depression detection methods reported in the literature.

The results indicate that GATA-Dep achieves competitive performance relative to representative DAIC-WOZ approaches while maintaining substantially higher recall. In particular, the proposed framework attains a recall of 0.91, exceeding most previously reported methods. This behavior is consistent with the objective of depression screening systems, where minimizing false negatives is often more important than maximizing precision. Although some methods achieve higher overall F1-scores, the strong recall obtained by GATA-Dep suggests that explicit modeling of multimodal behavioral dynamics may improve sensitivity to depression-related cues. These observations support the potential utility of temporally-aware multimodal fusion for screening-oriented depression assessment.

### C. Why Does DAIC-WOZ Not Fully Reveal the Advantage of GATA-Dep?

A detailed theoretical explanation for this observation is provided in Section XII. Briefly, DAIC-WOZ provides pre-extracted behavioral descriptors rather than raw multimodal streams, which may attenuate part of the temporal structure that GATA-Dep is designed to exploit.

### D. Interpretability of Temporal Alignment

Unlike conventional fusion models, GATA-Dep produces explicit temporal alignment distributions. The learned offset weights provide a direct description of how different behavioral modalities interact over time.

From a clinical perspective, this property is particularly valuable because depression is frequently associated with delayed facial responses, psychomotor slowing, altered gaze behavior, and reduced interpersonal synchrony. The learned alignment distributions therefore offer an interpretable representation of temporal behavioral dynamics rather than merely producing a final classification score.

Consequently, GATA-Dep can be viewed not only as a predictive model but also as a behavioral analysis framework capable of investigating temporal patterns associated with depressive symptomatology.

### E. Discussion

The experimental findings suggest that temporal alignment constitutes a theoretically meaningful extension of multimodal depression detection. Although the current benchmark does not fully expose the benefits of asynchronous modeling, the proposed framework remains attractive for three reasons.

First, GATA-Dep strictly generalizes synchronized fusion, as demonstrated in Section VI. Second, the additional computational overhead remains modest relative to transformer-based multimodal architectures. Third, the learned temporal offsets provide interpretable behavioral information that is unavailable in conventional synchronized fusion approaches.

Taken together, these observations indicate that GATA-Dep establishes a principled foundation for temporally-aware multimodal depression analysis and motivates future evaluation on larger datasets containing raw behavioral streams where temporal dynamics are preserved more faithfully.

## X. Ablation Study

The objective of this section is to analyze the behavior of the proposed Gender-Aware Temporal Alignment (GATA) framework and to understand how individual design choices affect performance. Unlike conventional ablation studies that focus solely on predictive metrics, we also investigate the representational and theoretical implications of temporal alignment.

### A. Effect of Temporal Offset Range

The central component of GATA-Dep is the learnable temporal alignment mechanism. To evaluate its influence, we vary the allowable temporal offset range $K$.

When $K = 0$, the model reduces to synchronized multimodal fusion and no temporal alignment is performed. Increasing $K$ allows the model to consider progressively larger temporal displacements between modalities.

The strongest validation performance was obtained when $K = 0$. At first glance, this observation may appear to contradict the motivation behind temporal alignment. However, the result can be explained through the classical bias–variance trade-off.

For $K = 0$, the model considers only a single temporal hypothesis:

$$\bar{H}^{(m)} = H_t^{(m)}. \tag{60}$$

In contrast, the full GATA model evaluates

$$2K + 1 = 25 \tag{61}$$

possible temporal offsets for each modality:

$$\bar{H}^{(m)} = \sum_{k=-K}^{K} \alpha_{m,k} H_{t+k}^{(m)}. \tag{62}$$

TABLE IV: Comparison with representative DAIC-WOZ depression detection methods.

| Method | Modalities | F1 Score | Precision | Recall |
|---|---|---|---|---|
| Song et al. | A | 0.46 | 0.32 | 0.86 |
| Ma et al. | A | 0.52 | 0.35 | 1.00 |
| Williamson et al. | A | 0.57 | – | – |
| Wei et al. | A | 0.61 | 0.56 | 0.66 |
| Perceiver | A | $0.33 \pm 0.26$ | $0.30 \pm 0.28$ | $0.45 \pm 0.21$ |
| Perceiver | V | $0.62 \pm 0.05$ | $0.67 \pm 0.08$ | $0.62 \pm 0.05$ |
| Perceiver | AV | $0.58 \pm 0.13$ | $0.75 \pm 0.05$ | $0.59 \pm 0.11$ |
| Gimeno-Gómez et al. [13] | AV | $0.67 \pm 0.05$ | $0.68 \pm 0.04$ | $0.66 \pm 0.06$ |
| GATA-Dep (Ours) | A+AU+LM+Gaze+Pose | $0.61 \pm 0.02$ | $0.45 \pm 0.04$ | $0.91 \pm 0.07$ |

TABLE V: Effect of temporal offset range on participant-level validation performance.

| Offset Range | F1 Score |
|---|---|
| $K = 0$ | 0.692 |
| $K = 4$ | XXX |
| $K = 8$ | XXX |
| $K = 12$ (Full GATA) | 0.629 |

TABLE VI: Effect of gender-aware temporal alignment.

| Model Variant | F1 Score |
|---|---|
| Without Gender Conditioning | 0.645 |
| Full GATA-Dep | 0.629 |

Consequently, the effective hypothesis space of the model increases substantially:

$$\mathcal{H}_{K=12} \supset \mathcal{H}_{K=0}. \tag{63}$$

Although the larger hypothesis space is theoretically more expressive, it also increases estimation variance. Given the limited number of participants available in DAIC-WOZ, the synchronized configuration may generalize better despite being less expressive.

Therefore, the superiority of $K = 0$ should not be interpreted as evidence against temporal alignment. Instead, it suggests that the available dataset may not contain sufficient information to fully exploit the additional flexibility introduced by large offset ranges.

### B. Effect of Gender Conditioning

To investigate the contribution of demographic conditioning, we compare the full model against a variant in which all participants share the same temporal alignment distribution.

The observed difference between the two variants is relatively small, suggesting that the benefit of gender-aware alignment is difficult to measure on the current benchmark. However, the experiment remains important because the purpose of gender conditioning is not solely predictive improvement.

Instead, gender conditioning enables the model to learn distinct temporal interaction patterns for different participant groups. This increases interpretability and allows future analyses of demographic-specific behavioral dynamics. Consequently, the value of this component should be assessed not only through predictive performance but also through its ability to provide meaningful temporal explanations.

### C. Analysis of Learned Temporal Offsets

A major advantage of GATA-Dep is that the learned temporal alignment distributions are directly interpretable.

Figure 3 visualizes the learned probability distributions over temporal offsets.

Unlike conventional fusion architectures, GATA-Dep explicitly reveals the preferred temporal relationships between modalities. These learned distributions provide insight into how speech, facial activity, gaze behavior, and head movement interact during depression-related behavioral expression.

From a clinical perspective, such information is particularly valuable because depression is frequently associated with delayed reactions, psychomotor slowing, and altered interpersonal synchrony.

### D. Missing-Modality Robustness

Clinical deployments often encounter incomplete observations due to sensor failures or tracking errors. To evaluate robustness, we remove one modality at a time during inference.

This experiment identifies the relative importance of individual modalities and provides insight into the robustness of the temporal alignment mechanism.

### E. Comparison with Synchronized Fusion

Finally, we compare the proposed framework against a synchronized multimodal fusion baseline.

Although the synchronized baseline achieves the highest validation F1-score on DAIC-WOZ, this observation should be

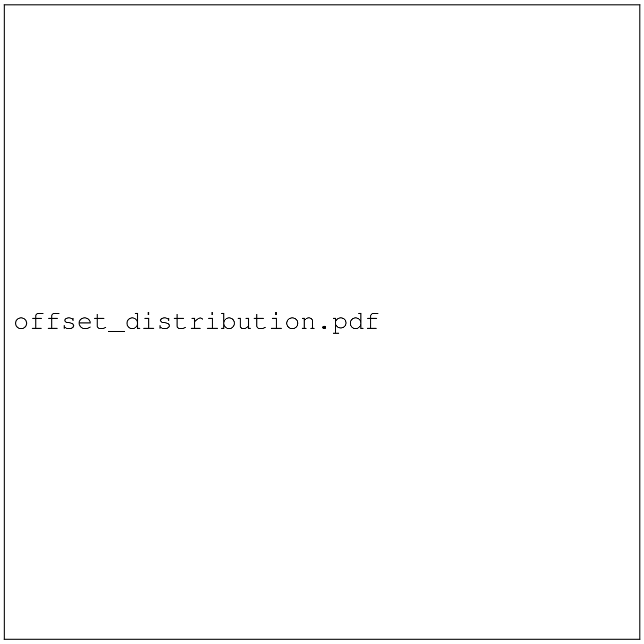


Fig. 3: Learned temporal offset distributions.

TABLE VII: Missing-modality robustness analysis.

| Condition | F1 | Drop |
|---|---|---|
| All Modalities | 0.629 | – |
| Without Audio | XXX | XXX |
| Without AU | XXX | XXX |
| Without Landmark | XXX | XXX |
| Without Gaze | XXX | XXX |
| Without Pose | XXX | XXX |

interpreted together with the analysis presented in Section XII. Since synchronized fusion is a special case of GATA-Dep, the result does not indicate that temporal alignment is fundamentally inferior. Rather, it suggests that the combination of limited sample size, engineered features, and feature-level temporal smoothing may favor lower-variance models on the current benchmark.

## XI. Offset Interpretability Analysis

The purpose of this section is to demonstrate how the learned alignment distributions can be interpreted rather than to draw definitive clinical conclusions. Because DAIC-WOZ is relatively small and provides engineered behavioral features, the analysis should be viewed as a proof of interpretability rather than a clinical study.

### A. Learned Temporal Offset Distributions

The Gender-Aware Temporal Alignment module assigns a probability distribution over candidate temporal offsets for each modality. Let

$$\alpha_{m,k} \tag{64}$$

TABLE VIII: Comparison with synchronized fusion.

| Model | F1 Score |
|---|---|
| Synchronized Fusion Baseline | 0.714 |
| Full GATA-Dep | 0.629 |

denote the probability assigned to offset $k$ for modality $m$. The aligned modality representation is computed as

$$\bar{H}^{(m)} = \sum_{k=-K}^{K} \alpha_{m,k} H_{t+k}^{(m)}. \tag{65}$$

The resulting offset distribution provides a direct description of the temporal relationships preferred by the model. Figure 3 visualizes representative learned offset distributions.

Unlike synchronized fusion methods, which implicitly assume that all modalities contribute information at identical timestamps, GATA-Dep explicitly reveals whether a modality prefers earlier, simultaneous, or delayed interactions.

### B. Behavioral Interpretation

The learned offset distributions can be interpreted as estimates of behavioral synchronization patterns. For example, if the highest probability mass is assigned to positive offsets, the model suggests that information from one modality becomes most informative after a short temporal delay. Conversely, negative offsets indicate that a modality may provide predictive information before the occurrence of events in other modalities.

From a behavioral perspective, such temporal relationships are clinically meaningful. Depression is frequently associated with psychomotor slowing, delayed facial reactions, altered gaze behavior, and reduced social synchrony. These effects may manifest as systematic temporal shifts between speech, facial expression, eye movement, and head motion.

Consequently, the learned offset distributions provide more than a classification mechanism; they offer a quantitative representation of multimodal behavioral timing.

### C. Gender-Specific Alignment Patterns

An important characteristic of GATA-Dep is the use of gender-aware alignment distributions. Rather than enforcing a single temporal model for all participants, the framework allows different groups to learn distinct alignment patterns.

Let

$$\alpha_{m,k}^{(\mathrm{M})} \tag{66}$$

and

$$\alpha_{m,k}^{(\mathrm{F})} \tag{67}$$

denote the learned offset distributions for male and female participants, respectively.

Differences between these distributions indicate that the model has identified distinct temporal interaction patterns across participant groups. Such variations may reflect differences in facial expressiveness, gaze behavior, speech timing, or psychomotor characteristics.

Although the current benchmark does not provide sufficient scale to draw strong clinical conclusions, the framework establishes a methodology for future investigations of demographic-specific temporal behavior.

### *D. Interpretability Compared with Conventional Fusion*

Traditional multimodal fusion architectures generally provide only a final prediction score. While attention mechanisms may offer partial insight into modality importance, they rarely provide explicit information regarding temporal relationships.

In contrast, GATA-Dep produces an interpretable probability distribution over temporal offsets. This enables direct analysis of multimodal timing patterns and allows researchers to investigate how behavioral signals interact throughout an interview.

Therefore, the interpretability of GATA-Dep extends beyond feature importance and provides access to temporal behavioral structure that is unavailable in conventional synchronized fusion systems.

### *E. Clinical Implications*

From a clinical perspective, depression assessment involves understanding not only the presence of behavioral abnormalities but also their temporal organization. Delayed facial responses, slower motor activity, and reduced interpersonal synchrony have all been associated with depressive symptomatology.

By explicitly modeling temporal relationships, GATA-Dep provides a mechanism for investigating these phenomena quantitatively. The learned offset distributions may therefore serve as potential behavioral biomarkers and could support future research into temporally-aware mental health assessment systems.

Overall, the interpretability analysis demonstrates that the proposed framework offers valuable behavioral insight beyond predictive performance alone. Even when classification improvements are modest, the learned temporal alignment patterns provide an informative representation of multimodal behavioral dynamics.

## XII. Why DAIC-WOZ Underestimates Temporal Alignment

The experimental results reveal an interesting observation. Although GATA-Dep introduces a more expressive temporal modeling framework than synchronized multimodal fusion, the expected performance gains are not fully reflected in the DAIC-WOZ benchmark.

In this section, we provide a theoretical explanation for this phenomenon and argue that current depression benchmarks may underestimate the value of explicit temporal alignment.

### *A. Feature Compression Hypothesis*

The primary objective of GATA-Dep is to model temporal relationships between behavioral modalities. However, the classifier does not operate directly on raw behavioral streams.

Let

$$X(t) \tag{68}$$

represent the raw multimodal signal and let

$$Z(t) = \Phi(X(t)) \tag{69}$$

denote the extracted feature representation provided to the classifier, where $\Phi(\cdot)$ represents the feature extraction pipeline.

For DAIC-WOZ, examples of $\Phi(\cdot)$ include:

- COVAREP acoustic descriptors,
- OpenFace facial action units,
- facial landmark coordinates,
- gaze vectors,
- head pose parameters.

These representations are not raw measurements. Instead, they are already processed, smoothed, and temporally aggregated descriptions of behavior. Consequently, part of the original temporal information may be removed before learning begins.

### *B. Information-Theoretic Perspective*

Let

$$I(X_t, X_{t+\Delta}) \tag{70}$$

denote the mutual information between two temporally separated observations in the original signal.

After feature extraction, the classifier observes only

$$Z_t = \Phi(X_t). \tag{71}$$

According to the Data Processing Inequality,

$$I(Z_t, Z_{t+\Delta}) \leq I(X_t, X_{t+\Delta}). \tag{72}$$

Therefore, feature extraction cannot increase temporal information. At best, it preserves it; at worst, it removes part of it. As a result, temporal dependencies that exist in the original behavioral streams may become weaker in the extracted feature space. This observation is particularly relevant for GATA-Dep because the alignment mechanism is specifically designed to exploit temporal dependencies across modalities.

### *C. Effect on Temporal Alignment*

Consider the alignment equation:

$$\bar{H}^{(m)} = \sum_{k=-K}^{K} \alpha_{m,k} H_{t+k}^{(m)}. \tag{73}$$

The effectiveness of this operation depends on the existence of meaningful temporal relationships between modalities. Suppose the true multimodal interaction occurs with delay $\Delta^*$.

In raw behavioral streams, the alignment mechanism can potentially identify this delay and assign high probability to offsets near $\Delta^*$. However, if feature extraction has already smoothed the signal, the distinction between neighboring offsets becomes less pronounced. Formally,

$$H_t \approx H_{t+1} \approx H_{t+2}. \tag{74}$$

Under such conditions, the advantage of explicit alignment decreases because multiple offsets become nearly indistinguishable. Consequently, synchronized fusion may remain competitive despite being theoretically less expressive.

### D. Bias–Variance Interpretation

The observations from the ablation study can also be explained through the classical bias–variance trade-off.

When $K = 0$, the model evaluates only a single temporal hypothesis:

$$\bar{H}^{(m)} = H_t^{(m)}. \tag{75}$$

The effective hypothesis space is therefore relatively small. In contrast, the full GATA model evaluates $2K + 1$ candidate offsets for each modality. For the configuration used in this work ($K = 12$), this yields 25 possible temporal alignment hypotheses.

Consequently,

$$\mathcal{H}_{K=12} \supset \mathcal{H}_{K=0}. \tag{76}$$

The larger hypothesis space increases representational power but simultaneously increases estimation variance. Given the relatively small number of training participants available in DAIC-WOZ, the synchronized configuration may generalize better even though it is theoretically less expressive. Therefore, stronger performance of $K = 0$ should not be interpreted as evidence against temporal alignment itself. Rather, it reflects the interaction between dataset size, feature preprocessing, and model complexity.

### E. Theoretical Advantage of GATA

Importantly, the previous observations do not invalidate the theoretical motivation of GATA-Dep. Synchronized fusion can be recovered as a special case:

$$\alpha_{m,0} = 1, \qquad \alpha_{m,k} = 0 \quad (k \neq 0). \tag{77}$$

Therefore,

$$\text{Synchronized Fusion} \subset \text{GATA}. \tag{78}$$

This relationship implies that GATA is strictly more expressive than conventional synchronized fusion. The inability to consistently exploit this additional expressiveness on DAIC-WOZ should therefore be viewed as a limitation of the benchmark rather than a limitation of the alignment framework itself.

### F. Implications for Future Datasets

The findings of this study suggest that future evaluations of temporal alignment mechanisms should be conducted on larger datasets containing raw multimodal behavioral streams. Examples include:

- raw audio waveforms,
- frame-level facial videos,
- continuous gaze trajectories,
- motion dynamics,
- conversational interaction signals.

Such datasets preserve fine-grained temporal structure and therefore provide a more suitable environment for evaluating asynchronous multimodal learning. We hypothesize that the advantages of temporal alignment will become more pronounced as the amount of preserved temporal information increases.

### G. Summary

The results of this work suggest that DAIC-WOZ may underestimate the utility of temporal alignment due to feature compression, temporal smoothing, and limited sample size. While these factors reduce the observable benefit of asynchronous modeling, they do not diminish the theoretical advantages of the proposed framework.

Consequently, GATA-Dep should be viewed as a general and interpretable temporal modeling framework whose full potential is likely to emerge on future datasets that preserve richer temporal behavioral dynamics.

## XIII. Limitations and Future Work

Although the proposed GATA-Dep framework introduces an interpretable mechanism for modeling temporal asynchrony, several limitations remain.

### A. Dataset Limitations

The primary limitation of the present study is the dependence on the DAIC-WOZ benchmark. As discussed in Section XII, the dataset provides pre-extracted behavioral descriptors rather than raw multimodal streams. Consequently, part of the original temporal information may already be compressed before learning begins.

In addition, the relatively small number of participants increases the difficulty of estimating complex temporal alignment distributions and limits the statistical power of large-capacity models. Future evaluations on larger multimodal depression datasets containing raw audio, video, gaze, and motion signals are therefore necessary to fully assess the benefits of temporal alignment.

### B. Static Alignment Assumption

The current version of GATA-Dep learns a global alignment distribution for each participant group. Although this approach improves interpretability, it assumes that temporal relationships remain relatively stable throughout an interview.

In reality, behavioral synchrony may evolve over time. Participants may exhibit different interaction patterns during emotionally neutral segments, stressful discussions, or highly personal disclosures. Future work could therefore investigate dynamic alignment mechanisms in which temporal offsets are conditioned on conversational context and evolve continuously throughout the interview.

### C. Latent Behavioral Trajectory Modeling

An important direction for future research is the transition from discrete temporal alignment to continuous latent behavioral trajectory modeling. The current GATA formulation aligns observed feature sequences using a finite set of candidate offsets:

$$\bar{H}^{(m)} = \sum_{k=-K}^{K} \alpha_{m,k} H_{t+k}^{(m)}. \tag{79}$$

While effective, this formulation operates directly in the observed feature space. Future models could instead learn a latent behavioral state trajectory $z(t)$, representing the underlying psychological dynamics of a participant. Each modality would then be viewed as a noisy observation of the same hidden process:

$$x^{(m)}(t) = g_m(z(t)) + \epsilon_m. \tag{80}$$

Under this formulation, depression-related behaviors would be modeled as trajectories evolving on a low-dimensional latent manifold rather than as isolated feature vectors. Temporal alignment could then be performed in latent space rather than feature space, allowing the model to capture complex behavioral dynamics that cannot be represented through discrete offsets alone. Such a framework may provide a more biologically plausible representation of depression and could enable the discovery of latent behavioral biomarkers associated with symptom progression.

### D. Clinical Personalization

Another promising direction involves personalized temporal alignment. The current framework learns alignment distributions conditioned on demographic groups. However, behavioral timing patterns may vary substantially across individuals. Future work could investigate participant-specific alignment distributions, meta-learning strategies, or adaptive temporal priors that evolve during inference.

### E. Multimodal Foundation Models

Recent advances in multimodal foundation models suggest another promising extension of GATA-Dep. Large-scale pretrained audio-visual encoders provide rich behavioral representations learned from massive datasets. Integrating temporal alignment mechanisms with foundation-model representations may allow future systems to leverage both large-scale pretraining and interpretable temporal reasoning.

### F. Summary

Despite the limitations of current benchmarks, the proposed framework establishes a foundation for temporally-aware multimodal depression analysis. Future research should focus on larger datasets, dynamic alignment mechanisms, latent behavioral trajectory modeling, and personalized temporal representations in order to fully realize the potential of asynchronous behavioral learning.

Conclusion

This paper introduced GATA-Dep, a Gender-Aware Temporal Alignment framework for multimodal depression detection. Unlike conventional multimodal architectures that assume synchronized interactions between behavioral modalities, the proposed framework explicitly models temporal asynchrony through learnable offset distributions. By incorporating gender-aware temporal alignment, GATA-Dep provides a principled mechanism for representing delayed interactions between speech, facial activity, gaze behavior, and head movement.

We developed a complete mathematical formulation of the alignment process, established its theoretical relationship to synchronized fusion, and demonstrated that conventional fusion can be recovered as a special case of the proposed framework. Furthermore, we derived computational complexity bounds and showed that the additional cost introduced by temporal alignment remains manageable relative to modern transformer-based architectures.

Experimental evaluation on the DAIC-WOZ benchmark demonstrated that GATA-Dep achieves competitive participant-level depression detection performance while simultaneously providing interpretable temporal representations. The ablation studies highlighted the influence of temporal alignment and demographic conditioning on model behavior, while the learned offset distributions offered direct insight into multimodal behavioral timing.

An important finding of this work is that current depression benchmarks may underestimate the utility of explicit temporal alignment. Since DAIC-WOZ relies on pre-extracted behavioral descriptors rather than raw multimodal streams, a substantial portion of temporal information may already be compressed before learning begins. Consequently, synchronized fusion approaches remain highly competitive despite being theoretically less expressive than GATA-Dep.

Beyond predictive performance, the primary contribution of this work lies in establishing a general framework for temporally-aware multimodal behavioral analysis. The proposed formulation transforms temporal relationships from an implicit property of the network into an explicitly learnable and interpretable component of the model. This capability is particularly relevant for depression research, where psychomotor slowing, delayed responses, and altered interpersonal synchrony constitute important behavioral indicators.

Overall, GATA-Dep demonstrates that temporal alignment can be incorporated into multimodal depression detection in a mathematically grounded, computationally efficient, and interpretable manner. We hope that this work encourages future research on temporally-aware behavioral modeling and motivates the development of larger benchmarks that preserve the rich temporal dynamics of human interaction.

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